

Problem Set 1

Due at the beginning of class on Thu Jun 24th

Answers must be typed. Send solutions to raffaele.saggio@ubc.ca. You can work in teams under the constraint $\max(\text{TeamSize}) = 2$

Question 1: Returns to Schooling

Consider an infinitely-lived person making a choice about how long to go to school. She faces a discount rate r . If she chooses S years of schooling, she earns $y(S)$ from date S onward. While at school, she earns nothing in the labor market, but receives a scholarship payment. Specifically, if she attends S years of schooling, she is paid a transfer T from time 0 to time S . Her goal is to maximize the present value of earnings and scholarship payments.

1. Write down a continuous-time expression for the person's objective function. Derive and interpret a condition to characterize the optimal duration of schooling.
2. Suppose that agents differ in ability, α_i , and the human capital production function for individual i takes the form

$$y_i(S) = \exp(\alpha_i + \beta S) \quad (1)$$

Solve for the optimal duration of schooling. Do higher-ability agents get more or less schooling? Why?

3. Now suppose both tuition payments and ability differ across agents: the scholarship offered agent i is T_i . Derive the probability limit of the slope coefficient from a bivariate OLS regression of log earnings on schooling in terms of β , $\sigma_T^2 = \text{Var}(\log T_i)$, $\sigma_\alpha^2 = \text{Var}(\alpha_i)$, and $\sigma_{T\alpha} = \text{Cov}(\log T_i, \alpha_i)$. Discuss how the bias of OLS varies with these parameters.
4. Now suppose that $T_i = T$ (tuition payments are constant in the population) but $y_i(S) = \exp(\alpha_i + \beta_i S)$ (proportional returns to schooling differ across individuals). Derive the probability limit of the slope coefficient from a bivariate OLS regression of log earnings on schooling in terms of β , $\sigma_\beta^2 = \text{Var}(\beta_i)$, $\sigma_\alpha^2 = \text{Var}(\alpha_i)$, and $\sigma_{\beta,\alpha} = \text{Cov}(\beta_i, \alpha_i)$.

Question 2: Keane and Wolpin (1997)

This question asks you to solve and estimate a simplified version of the model from Keane and Wolpin (1997). An agent makes choices at two ages, $a \in \{1, 2\}$. At each age, she chooses between three alternatives: working in a white collar occupation, working in a blue collar occupation, and going to school. These alternatives are labeled $m \in \{1, 2, 3\}$. For alternatives 1 and 2, the flow utility of working at age a is

$$R_m(a) = \alpha_m + \beta_m G(a) + \gamma_m X_m(a) + \epsilon_m(a),$$

where $G(a)$ is education and $X_m(a)$ is experience in occupation m . The utility of going to school is

$$R_3(a) = \epsilon_3(a).$$

The agent enters period 1 with no experience or education. Assume that the $\epsilon_m(a)$ are iid draws from an extreme value type I distribution, independent over time. The $\epsilon_m(2)$ are unknown to the agent in period 1. The agent has a discount rate $\delta \in (0, 1)$.

1. Let $S(2)$ denote the vector of state variables entering age 2 (before the realization of $\epsilon_m(2)$). What are the elements of this vector? What is the probability that the agent chooses each alternative in period 2, as a function of $S(2)$?
2. Define the value function in period 2,

$$V_2(S(2)) = \max_{m \in \{1,2,3\}} R_m(2; S(2)).$$

Write down an expression for $E[V_2(S(2))]$ for each possible value of $S(2)$ prior to the realization of $\epsilon_m(2)$. (Hint: In a discrete choice problem where the utility of alternative k is $U_{ik} + \eta_{ik}$, where the η_{ik} are distributed iid extreme value type I, we have $E[\max_k U_{ik} + \eta_{ik}] = \nu + \log(\sum_k \exp(U_{ik}))$, where ν is a constant of integration that you can ignore.)

3. Use your results from part (2) to write down an expression for the agent's expected payoff from each choice in period 1. What is the *ex ante* probability of choosing each alternative in period 1, prior to the realization of $\epsilon_m(1)$?
4. Suppose you have data for a sample of agents choosing according to this model. Let $C_i(a) \in \{1, 2, 3\}$ denote individual i 's choice at age a . The observed data for individual i is the choice sequence $(C_i(1), C_i(2))$. Use your results from parts (1) and (3) to write down the likelihood that individual i chooses the sequence $(C_i(1), C_i(2))$. Take the log and sum over individuals to obtain an expression for the sample log-likelihood.
5. Download the data set "kw_example.csv" from the course website. This data set includes 5,000 simulated observations from the model. Import the data into a statistical software package of your choosing. Estimate the parameter vector

$$\theta \equiv (\alpha_1, \beta_1, \gamma_1, \alpha_2, \beta_2, \gamma_2, \delta)'$$

by maximum likelihood. (I recommend using Matlab and minimizing the negative of the log-likelihood using the *fminunc* routine. You may wish to search over $\log \delta$ rather than δ to keep the discount rate positive at every evaluation of the likelihood.) Attach your computer code, and make a table showing your parameter estimates; you may also provide standard errors for extra credit. Briefly provide some economic interpretation for your parameter estimates and discuss the tradeoffs faced by the agent when choosing an alternative in period 1.